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Paleomagnetic mapping of the regional fluid flow event that mineralized the Upper Mississippi Valley Zn–Pb ore district, Wisconsin, U.S.A.

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ABSTRACT

The Upper Mississippi Valley (UMV) zinc–lead district contains archetypal examples of Mississippi Valley-type (MVT) ore deposits. Specimens (332) from 25 sites in the Lower Ordovician Oneota dolomite were collected from around the northern and eastern sides of the UMV district on the Wisconsin Arch in southwestern Wisconsin. Most specimens yield one of two characteristic remanence (ChRM) components, A and B, in varying proportion at each site. Component B resides in magnetite mostly and gives a primary Early–Middle Ordovician paleopole at 15.1°N, 150.5°E ($\delta p = 2.4°$, $\delta m = 4.7°$), and component A resides mainly in pyrrhotite and gives an Early Permian remagnetization paleopole at 47.4°N, 127.3°E ($\delta p = 1.1°$, $\delta m = 2.2°$) that matches the UMV zinc–lead ore's paleopole. The percentage of A component specimens at each site is used to map the boundary of the Early Permian fluid flow that formed the UMV ore over a distance of >200 km. The map also delineates areas of intense pyrrhotite authigenesis created by sulphidic fluid flow within the UMV district, and locates potential conduits in the Oneota dolomite through which fluid flow was delivered from an adjacent basin.

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1. Introduction

Modern regional subsurface fluid flows can be mapped geochemically and the brine age dated radiometrically to obtain the flow rate (e.g. Musgrove and Banner, 1993; Rostron and Holmden, 2000). Theoretically, radiometric dating can be used to map a regional fluid flow event in the geologic record (e.g. Coveney et al., 2000), however, the time and cost of doing the necessary number of petrological and radiometric analyses is prohibitive at present. Paleomagnetic methods provide an alternative method, particularly in Phanerozoic rocks of geologic plates where the apparent polar wander path (APWP) is well defined such as in North America (Van der Voo, 1993; McElhinny and McFadden, 2000). For example, Jackson et al. (1988) reported paleomagnetic results from 21 sites along a ~750 km profile in the Devonian Onondaga Limestone across New York State and into Ontario, Canada. They found that the entire profile of sites carried a late Paleozoic remagnetization or characteristic remanent magnetization (ChRM) in authigenic magnetite, and they concluded that the authigenesis was caused by the introduction of brines during the Alleghenian Orogeny (McCabe et al., 1989). In this study we apply this concept to look for the boundary of the Early Permian fluid flow event that mineralized Middle Ordovician carbonates to form the Mississippi Valley-type (MVT) zinc–lead (Zn–Pb) deposits of the classic Upper Mississippi Valley (UMV) ore district of southwestern Wisconsin and adjacent Illinois and Iowa (Fig. 1).

The UMV district is the archetypal example of a MVT Zn–Pb district, being first mined in the early 1700s (Heyl, 1983). It was a major Zn–Pb source in America from the mid 1800s to mid 1900s and ceased production in the 1980s. The UMV district covers ~8000 km² in southwest Wisconsin, Illinois and Iowa (Fig. 1), including hundreds of large and small mining operations, with trace Zn–Pb mineralization extending reputedly well beyond the boundary (Heyl and West, 1982). The UMV ore occurs mostly in Middle Ordovician carbonates that were remagnetized on ore emplacement in the Early Permian (Pannalal et al., 2004). Suggested sources for the mineralizing fluids have been: 1) upward convection from underlying Precambrian heat-producing granites (e.g. Spirakis and Heyl, 1996); 2) northward fluid flow through the Illinois basin from the Ouachitan orogen (e.g. Rowan and Goldhaber, 1996; Rowan and de Marsily, 2001); and, 3) westward fluid flow through the Michigan basin from the Appalachian orogen (e.g. Bartos, 2001). Here we report paleomagnetic results from 25 sites distributed around the northern and eastern sides of the UMV district to test the above hypotheses.

2. Geology

The Paleozoic stratigraphic sequence in the UMV Zn–Pb district is shown in Fig. 2. If hydrothermal mineralizing fluids travelled through permeable basement fractures and/or the nonconformably overlying Upper Cambrian sandstones to reach the Middle Ordovician

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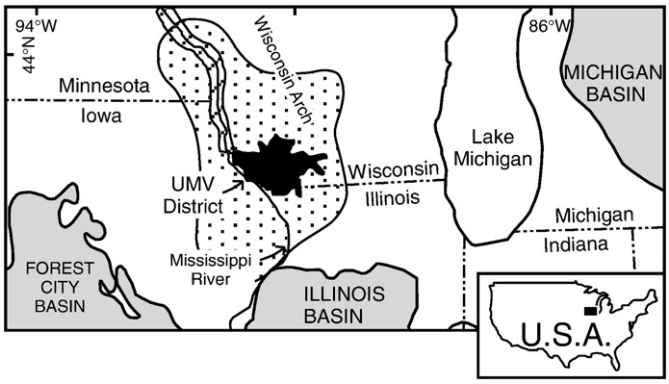


Fig. 1. Location of the Upper Mississippi Valley (UMV) MVT Zn–Pb district (black) in north central U.S.A., with its surrounding area of subeconomic mineralization (stippled), modified from Heyl and West (1982) and Pannalal et al. (2004). The hatching extending northwest from the UMV district along the Mississippi River valley depicts the region of Jackson and Van der Voo's (1985) paleomagnetic study of the Oneota Formation.

Platteville, Decorah, and Galena dolomites, limestones and shales that host the Zn–Pb mineralization, then the fluids must have passed through the intervening Prairie du Chien Group.

The Prairie du Chien Group is composed of the lower Oneota and upper Shakopee formations in the UMV district. The buff to sandy Oneota dolomite is known from conodonts to be lower Tremadocian (Smith et al., 1993) or 486 ± 3 Ma on the timescale of Gradstein et al. (2004). Also, the Oneota dolomite has given a ^{238}U–^{206}Pb dolomite age of 504 ± 14 Ma near site 1 (Fig. 3) that was apparently unaffected by the MVT mineralizing event (Winter and Johnson, 1995). The overlying Shakopee Formation is mid-Tremadocian or ~483 ± 2 Ma in age (Gradstein et al., 2004). Following deposition of the Prairie du Chien Group in a shallow tropical marine environment, there was prolonged subaerial exposure that produced extensive karst development and sinkhole-karst topography that caused extensive collapse breccias prior to Middle Ordovician sedimentation.

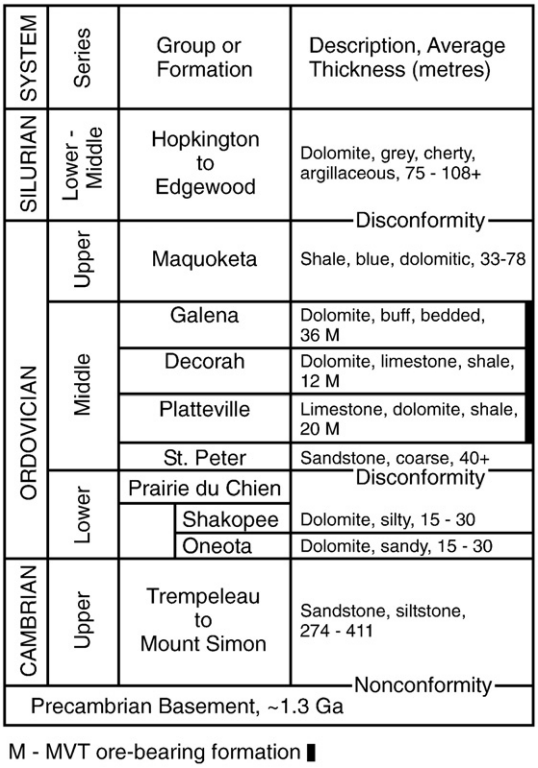


Fig. 2. Stratigraphy of the Upper Mississippi Valley zinc–lead district, modified from Heyl et al. (1973), Smith et al. (1993) and Pannalal et al. (2004).

Structurally, the Prairie du Chien Group is draped around the south end of the regional Wisconsin Arch (Fig. 1) with a regional dip of ~1°S to southwest and broad undulations with local dips of <3°. The strata thicken eastward, southward and southwestward into the intracratonic Michigan, Illinois and Forest City basins, respectively. Minor normal faults are common throughout the UMV district and the strata have a well-developed vertical joint system (Heyl et al., 1973).

The UMV MVT Zn–Pb mineralization occurs mostly as open-space filling in veins, solution and collapse breccias, joints, vugs, and in the pitches and flats of the minor faults. The mineralization has given Early Permian ages of 270 ± 4 Ma by the ^{87}Rb–^{86}Sr sphalerite method (Brannon et al., 1992) and of 282 ± 10 Ma by paleomagnetism (Pannalal et al., 2004). Since the carbonate host rocks give conodont Colour Alteration Indices (CAI) of 1.0 (Sangster et al., 1994), they have never been heated above ~60 °C (Symons and Cioppa, 2002, and references therein), which is consistent with known burial depths of ≤2500 m. However, the vein sphalerites have a median range of fluid inclusion homogenization temperatures of 115 ± 30 °C (Rowan and Goldhaber, 1996), implying that hydrothermal fluids with elevated temperatures must have ascended from depth through fractures and the Prairie du Chien Group to deposit the Zn–Pb mineralization.

3. Methods

Samples were collected from 25 sites in quarries and roadcuts of the Oneota Formation along a ~225 km corridor around the northern and eastern sides of the UMV district (Fig. 3). Six or seven oriented cores were collected at each site, yielding 11 to 17 standard paleomagnetic specimens per site and giving a collection of 332 specimens in total (Table 1). The specimens were stored in a magnetically-shielded room (ambient magnetic field of ~100γ) for ~2 years to allow their undesirable viscous remanent magnetization (VRM) components to substantially decay. Using a vertical automated 2G Enterprises 3-axis DC-SQUID magnetometer, the residual natural remanent magnetization (NRM) of each specimen was measured. Next, one typical specimen per site was alternating field (AF) demagnetized in 13 steps up to 120 mT using a Sapphire Instruments SI-4 demagnetizer and a second specimen was thermally demagnetized in 14 steps to 500 °C using a Magnetic Measurements MMTD-80 demagnetizer. Based on the results of these test specimens, most remaining specimens were thermally demagnetized in 8 steps from 225 to 380 °C to isolate the characteristic remanent magnetization (ChRM) carried by pyrrhotite that occurs preferentially with the sulphide mineralization in the UVM district (Pannalal et al., 2004), and that is used as the "tracer" for the mineralizing fluids in this study. Additional thermal steps were added as warranted. The few remaining glued specimens were AF demagnetized in 6 steps to 120 mT. Specimens from 9 sites were pulse magnetized in 14 steps to 900 mT using a Sapphire Instruments SI-6 pulse magnetizer and then AF demagnetized in 9 steps to 160 mT to examine their saturation isothermal remanent magnetization (SIRM) characteristics.

4. Results

The NRM intensities in 10^{-5} A/m for the Oneota specimens ranged from 0.06 to 822 with a median of 13.9 and lower and upper quartiles of 4.84 and 34.3, respectively.

Magnetization components were initially identified using contoured vector density plots (Kamb, 1959) for each AF and thermal demagnetization step. The NRM plot featured two concentrations of remanence directions. One concentration was directed steeply downward to the north and records modern weathering in the present Earth's magnetic field (PEMF) direction (Fig. 4a). The PEMF component resides primarily in goethite, identified by rapid thermal demagnetization by ~130 °C (Fig. 4c), and in hematite, both of which AF demagnetization fails to

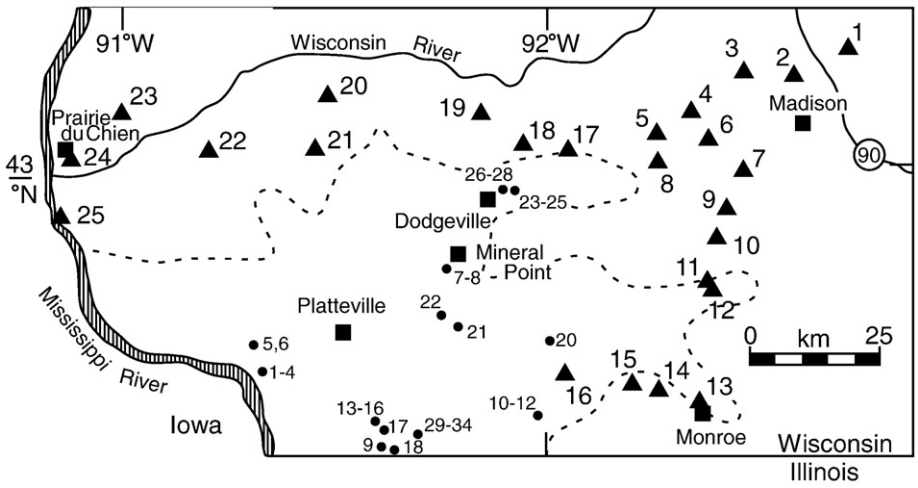


Fig. 3. Southwestern Wisconsin showing the core area of the UMV Zn–Pb district of Heyl et al. (1973) bounded by the dashed line, the Lower Ordovician Oneota Fm. sites of this study (▲23) and the zinc–lead mineralization and Middle Ordovician host rock sites of Pannalal et al. (2004) (●9).

demagnetize (Fig. 4a). The PEMF component also resides in pyrrhotite and magnetite as a viscous remanent magnetization (VRM) that is identified by the progressive directional movement away from the PEMF direction on both thermal and AF demagnetization (Fig. 4b). The hematite PEMF component is isolated only by thermal demagnetization at temperatures of >600 °C, which yields the PEMF direction either consistently or at high temperatures as a residual remanence with intensities of $<5\times10^{-6}$ A/m at the highest AF and thermal demagnetization steps. The PEMF component was dominant in 36 specimens (Table 1) but is not considered further. The second directional concentration on the Kamb plot was in a SSE and shallowly upward direction, and it is termed the A characteristic remanent magnetization (ChRM) component (Fig. 5). This directional concentration increases in intensity on a Kamb plot up to the ~250 °C to ~330 °C range whereupon it quickly dies out on thermal step demagnetization, indicating that it is carried mainly by pyrrhotite but also by minor magnetite (Fig. 5a–c). When present, the A component is isolated easily by both thermal (Fig. 5a–c) and AF step demagnetization (Fig. 5d), usually after the removal of a minor VRM component in the first couple of demagnetization steps. At the highest AF and thermal demagnetization steps, a second ChRM component was observed in some specimens in the ESE and shallowly downward direction (Fig. 6b) or its antiparallel WNW and shallowly upward direction (Fig. 6a), which is termed the B ChRM component. This component is carried mostly by magnetite, and is best isolated above 330 °C but before ~400 to ~450 °C when sulphides such as pyrite and pyrrhotite in the specimens start to oxidize in the oven to generate new magnetite with more intense random directions.

The SIRM acquisition (Fig. 7a) and decay (Fig. 7b) curves show the mixed magnetic mineralogy found in the Oneota dolomite specimens. Three specimens (6, 7 and 15 in Fig. 7) have acquisition curves that reach saturation ($0.9\times J_{900}$) by ~300 mT and subsequent decay curves that are typical of single domain pyrrhotite and/or magnetite. Another three curves (2, 8 and 22 in Fig. 7) have acquisition and decay curves suggesting that goethite and/or hematite are the main magnetic minerals present. The remaining three specimens (10, 18 and 21 in Fig. 6) have acquisition curves that rise rapidly up to ~300 mT and then gently ascend from ~300 to 900 mT, and their decay curves remove only about half of the SIRM intensity. These specimens contain a mixture of pyrrhotite and/or magnetite and goethite and/or hematite in variable proportions. The highly variable magnetic mineralogy found by the SIRM curves, as well as in the initial thermal and AF step demagnetization curves, led to the choice to use thermal step demagnetization on most of the Oneota specimens.

The specimen ChRM directions were isolated using orthogonal vector plots (Zijderveld, 1967) combined with the least-squares fitting method of Kirschvink (1980), except for a few specimens where the remagnetization circle method of Bailey and Halls (1984) was used. The specimen directions were averaged to get the site mean directions following Fisher (1953; Table 1). A number of specimens (34) gave either incoherent clustering of the step directions so that a site mean direction could not be defined or they defined directions that were anomalous relative to the site mean directions defined by the remaining specimens for the site (Table 1). Consequently these specimen directions were excluded from the site mean directions.

The A component site mean direction was defined in four or more specimens from 23 of the 25 sites (Table 1). Of the 23 sites, sites 5 and 19 had large cones of 95% confidence of 15.4° and 15.1°, respectively, are deemed to be insufficiently reliably clustered, and are excluded from the A component mean direction calculation. The remaining 21 sites give a mean direction of $D=155.7°$, $I=-12.6°$ ($N=21$, $\alpha_{95}=2.2°$, $k=212.7$) [declination ($D°$), inclination ($I°$), number of sites (N), radius of the cone of 95% confidence ($\alpha_{95}°$), and precision parameter (k) of Fisher (1953)] (Fig. 8).

After reversing WNW specimen ChRM directions to their antipolar ESE directions, the B component is dominant and well isolated in only four sites (19, 22, 23 and 24) (Table 1). These four sites give a mean direction $D=124.3°$, $I=15.9°$ ($N=4$, $\alpha_{95}=7.4°$, $k=157.1$). However, there are 13 other sites in which one to four specimens were found to retain the B component (Table 1) that together give a mean direction of $D=118.9°$, $I=23.4°$ ($N=13$, $\alpha_{95}=9.1°$, $k=21.6$) that is not significantly different at 95% confidence from the mean direction of the four well-isolated sites. In order to give each specimen in these 13 sites a comparable weighting relative to the specimens in the four sites, the 24 specimens in the 13 sites were grouped into 5 reconstituted sites based on their geographic location (Table 2). When these 5 reconstituted sites are grouped with the 4 original well-isolated B component sites (Table 1), they give a mean direction of $D=121.6°$, $I=19.2°$ ($N=9$, $\alpha_{95}=4.5°$, $k=131.7$) for the B component (Fig. 8).

5. Paleopoles

Using a mean site latitude of 43.0°N and longitude of 89.9°W (270.1°E), the A component mean ChRM direction of $D=155.7°$, $I=-12.6°$ ($\alpha_{95}=2.2°$) provides a paleopole at 47.4°N, 127.3°E ($\delta p=1.1°$, $\delta m=2.2°$) where δp and δm are the semi-axes of the oval of 95% confidence about the paleopole along, and perpendicular to, the site-paleopole great circle, respectively (Fisher, 1953) (Fig. 9a,b). The B component mean direction of $D=121.6°$, $I=19.2°$ ($\alpha_{95}=4.5°$) gives a paleopole at 15.1°N, 150.5°E ($\delta p=2.4°$, $\delta m=4.7°$) (Fig. 9a,c).

Table 1
Site mean remanence directions.

Site	Location		Specimens			Median J_0 intensity A/m × 10^{-5}	A remagnetization component direction						B primary component direction					
	Latitude °N	Longitude °W	me,	pe,	an		Specimens		Decl. °	Incl. °	α_{95} °	k	Specimens		Decl. °	Incl. °	α_{95} °	k
							ep,	rc					ep,	rc				
1	43.21	89.18	11,	0,	1	5.58	9,	1	161.2	−13.0	6.5	55.6						
2	43.16	89.33	13,	0,	0	22.8	12,	0	153.4	−10.1	5.6	60.3	1,	0	147.9	31.4		
3	43.19	89.40	13,	4,	2	50.8	6,	1	148.3	−11.3	10.2	35.9						
4	43.11	89.67	15,	3,	1	23.5	10,	0	152.3	−15.4	6.4	57.8	1,	0	108.1	16.3		
5	43.05	89.72	11,	4,	0	181	4,	0	144.1	0.9	15.4	36.7	3,	0	118.0	21.3	33.5	14.6
6	43.06	89.61	17,	4,	2	4.95	11,	0	161.8	−8.5	7.9	34.5						
7	42.99	89.63	11,	0,	0	19.4	10,	0	154.7	−16.5	6.1	63.7	1,	0	95.0	44.2		
8	43.00	89.71	11,	1,	0	12.7	9,	1	156.2	−14.5	5.5	77.8						
9	42.94	89.66	11,	3,	4	8.51	0,	0					4,	0	126.6	19.1	22.7	17.4
10	42.87	89.65	11,	0,	3	8.45	5,	1	158.7	−14.8	7.2	88.1	2,	0	130.7	29.9		9.0
11	42.81	89.69	13,	3,	3	33.0	5,	0	152.1	−13.1	8.2	87.6	2,	0	110.4	26.3		28.5
12	42.78	89.70	16,	0,	3	11.1	9,	4	153.3	−11.1	5.9	50.9						
13	42.63	89.69	11,	0,	0	18.1	8,	2	152.5	−15.0	6.9	50.1	1,	0	95.3	16.4		
14	42.63	89.76	15,	0,	0	18.9	14,	0	162.0	−11.0	5.3	58.2	1,	0	144.7	23.4		
15	42.63	89.81	16,	0,	3	36.8	13,	0	158.8	−13.7	4.9	71.9						
16	42.65	89.90	15,	2,	0	62.1	9,	4	154.3	−13.5	2.8	221.3						
17	43.01	89.85	15,	2,	1	6.12	12,	0	159.4	−13.5	6.2	49.3						
18	43.04	90.05	14,	0,	0	4.78	11,	1	147.4	−18.0	5.7	58.2	2,	0	121.8	3.5		68.1
19	43.11	90.12	14,	0,	1	1.78	3,	2	150.2	−6.9	15.1	26.6	6,	2	124.7	14.4	7.0	63.0
20	43.14	90.51	13,	8,	3	3.10	0,	0					2,	0	122.3	21.6		14.4
21	43.05	90.54	10,	0,	2	11.2	4,	2	150.4	−14.8	6.3	115.7	2,	0	119.0	18.2		26.5
22	43.02	90.77	15,	1,	3	7.89	2,	3	160.1	−6.3	6.7	132.9	7,	0	129.3	17.4	11.7	27.4
23	43.11	91.00	15,	0,	2	7.29	3,	1	153.2	−4.6	9.2	100.4	7,	2	115.0	17.4	9.5	30.2
24	43.04	91.13	16,	0,	0	5.34	7,	0	162.9	−9.8	11.5	28.6	9,	0	128.1	14.2	9.9	28.3
25	42.94	91.14	10,	1,	0	15.6	7,	0	156.9	−14.5	10.5	34.1	2,	0	105.0	22.5		18.1

Notes. Specimens: me — measured, pe — present Earth's magnetic field direction, an — anomalous direction; median: J_0 — median natural remanent magnetization intensity in amperes/meter (A/m); numbers: ep — end point direction, rc — remagnetization circle direction; decl. — declination in degrees (°); incl. — inclination; α_{95} — radius of cone of 95% confidence; k — precision parameter of Fisher (1953).

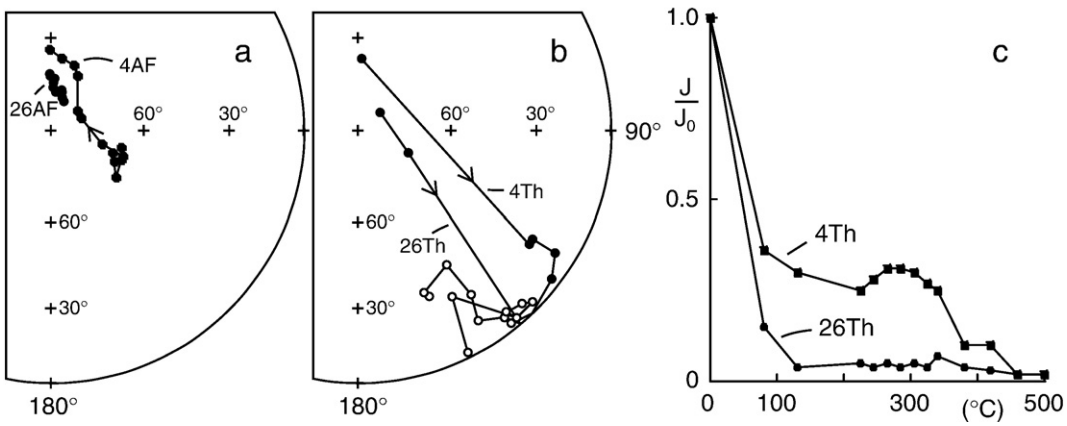


Fig. 4. Equal-area stereonet of the southeast quadrant showing for a specimen from site: 26 AF, the present Earth's magnetic field (PEMF) direction in goethite unaffected by AF demagnetization (Fig. 4a); 26 Th, the rapid demagnetization of goethite by thermal demagnetization and then the migration of the remanence direction in pyrrhotite from the PEMF to the A direction (Fig. 4bc); 4 Th, the initial rapid demagnetization of goethite by ~120 °C, of pyrrhotite by ~320 °C, and then of magnetite by ~460 °C (Fig. 4b and c); and, 4 AF, demagnetization of the mixed pyrrhotite–magnetite remanence to leave a residual PEMF direction in hematite (Fig. 4a). Solid (open) directions are in the lower (upper) hemisphere. The decay of intensity is expressed as a ratio of the NRM intensity (J/J_0) for thermal demagnetization. AF demagnetization does not affect substantially the remanence in either goethite or hematite.

6. Discussion

6.1. B component

The geologic interpretation of the B paleopole is straight forward. It falls on the combined North American and European apparent polar wander path (APWP) in North American co-ordinates of Van der Voo (1993) close to the reference paleopole for the Early and Middle Ordovician (Fig. 9c). Because the B component paleopole is close to the APWP age that matches its Lower Ordovician stratigraphic age, the B component is deemed to be a primary chemical remanent magnetization in magnetite that was acquired either at the time of

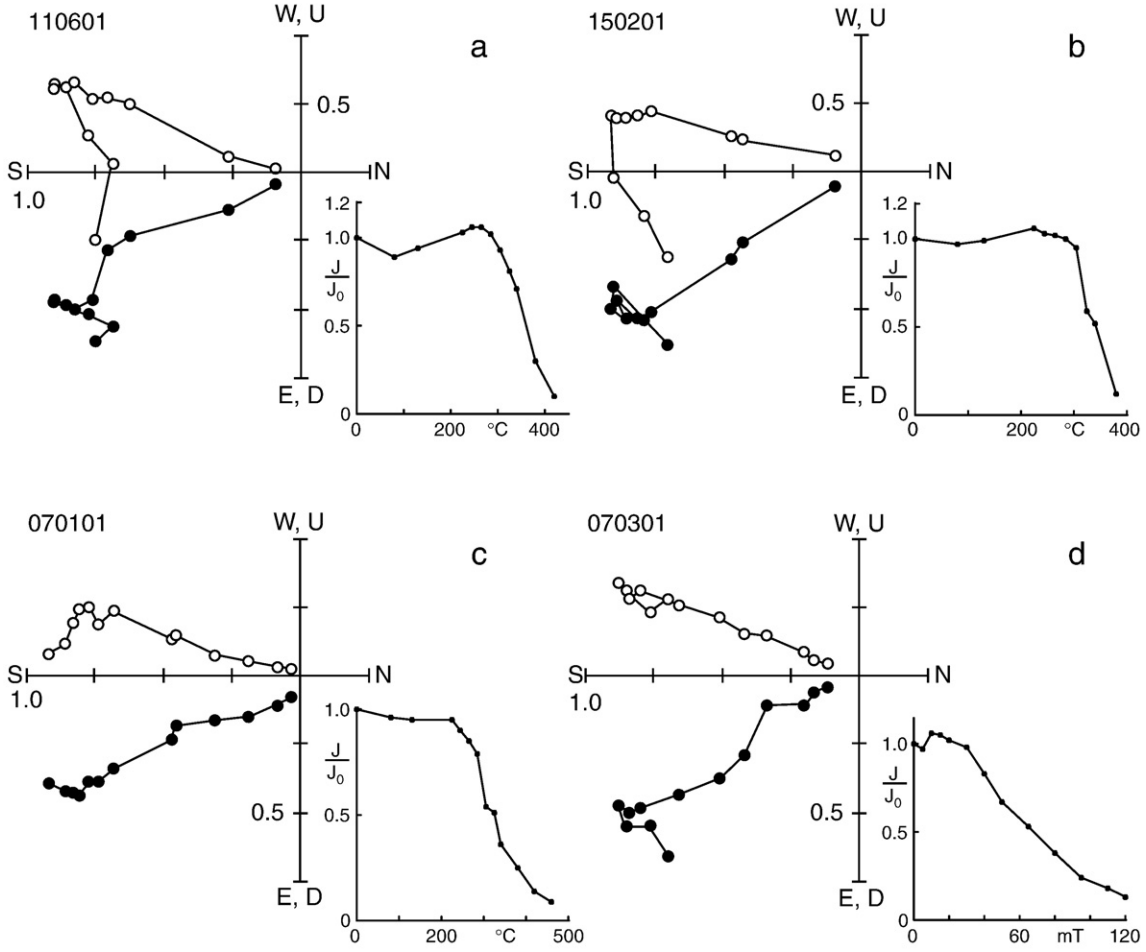


Fig. 5. Vector plots for the A component showing the specimen declination (solid circles) in the north–east–south–west (N–E–S–W) plane and the inclination (open circles) in the north–down–south–up (N–D–S–U) plane. Axial intensities (J) are given as a ratio of the NRM intensity (J_0). The intensity ratio is also plotted against temperature (°C) in panels a, b and c or against AF demagnetizing field (mT) in panel d. The specimen is identified by its six digit number (e.g. 110601).

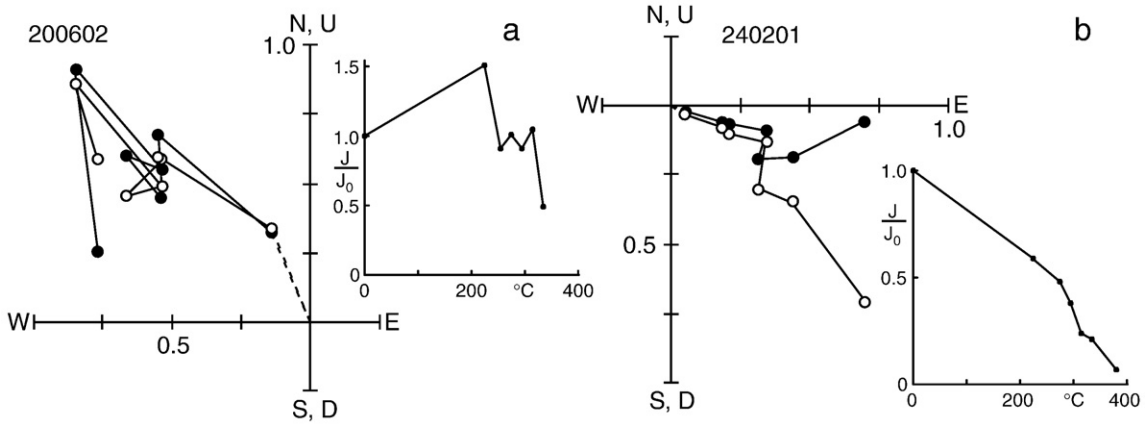


Fig. 6. Vector plot for: a) a normal B component; and, b) a reversed B component. Conventions as in Fig. 5.

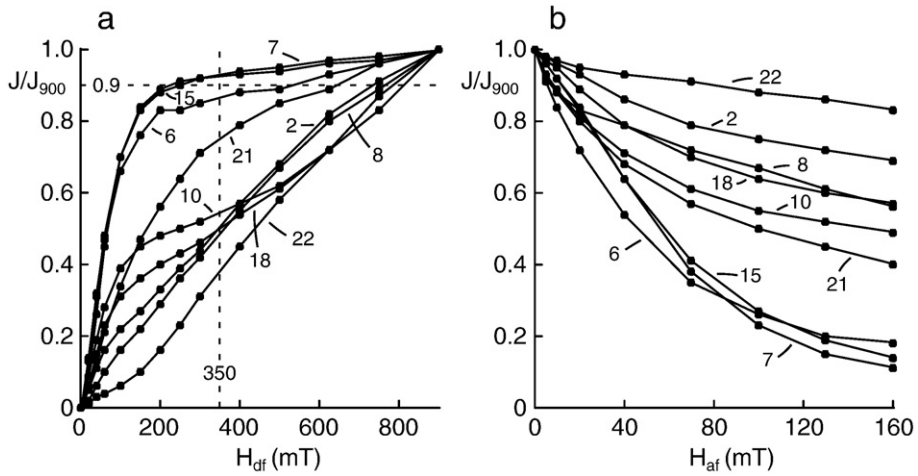


Fig. 7. Saturation isothermal remanent magnetization (SIRM) curves for nine representative specimens showing: a) acquisition of SIRM in a direct field (H_{df}), and b) demagnetization of SIRM in an alternating field (H_{af}) in millitesla (mT). J/J_{900} is the ratio of the measured remanence intensity (J) to the saturation remanence intensity (J_{900}). The dashed lines in (a) are at $J/J_{900} = 0.9$ and at $H_{df} = 350$ mT.

deposition or during diagenesis within a few million years after deposition.

Jackson and Van der Voo (1985) reported results from a previous paleomagnetic study of the Oneota Formation. They analyzed dolomite specimens from 17 sites that extend northwestward from Prairie du Chien, Wisconsin, for ~275 km along the Mississippi River valley in Wisconsin, Iowa and Minnesota (Fig. 1). As in this study, they found complex magnetization components in many specimens but were able to isolate a cluster of normal and reversed ChRM directions in magnetite in 10 sites from 32 end points and 41 remagnetization circle points that gave a paleopole at 10.4°N, 166.4°E ($N = 10$, $\alpha_{95} = 7.0°$, $k = 48.0$; Fig. 9c). Further, Jackson and Van der Voo (1985) reported that a paleomagnetic conglomerate test on six Oneota clasts in the overlying Lower Ordovician New Richmond sandstone was inconclusive but favoured the interpretation that the Oneota Formation's ChRM predates deposition of the sandstone. We attribute the difference between the Oneota paleopole found by Jackson and Van der Voo (1985) and found in this study to either the difficulty in truly isolating the B component from the complex magnetization components found in these strata or to lateral variation with time within the unit. Nevertheless, both paleopoles clearly indicate the preservation of a primary Early to Middle Ordovician remanence (Fig. 9c).

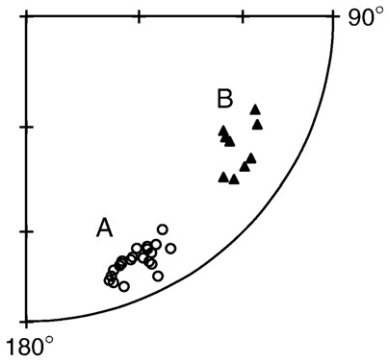


Fig. 8. Site mean remanence directions for A (circles) and B (triangles) components of the Oneota dolomite on the southeast quadrant of an equal-area stereonet. Downward and upward directions are shown as solid and open symbols, respectively.

Table 2
Reconstituted site mean directions for component B.

Sites	Specimens	Decl. °	Incl. °	α_{95}°	k
2, 4, 5	5	121.4	22.9	20.4	15.0
7, 9	5	121.5	24.7	22.9	12.1
10, 11, 13, 14	6	119.9	26.3	18.8	13.6
18, 20	4	122.3	14.1	17.5	27.5
21, 25	4	112.1	20.4	18.9	24.7

Note. Abbreviations as in Table 1.

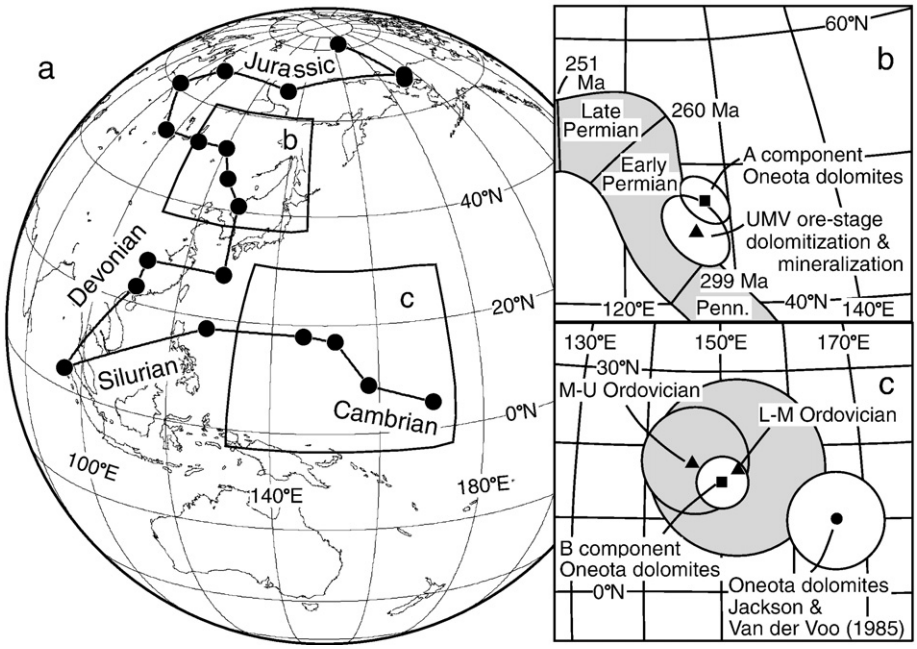


Fig. 9. a) Apparent polar wander path (APWP) of Van der Voo (1993) with insets showing the locations of enlargements of: b) the A component paleopoles for the Oneota dolomite of this study (square) and for the Upper Mississippi Valley district mineralization and dolomites of Pannalal et al. (2004; triangle) relative to the late Paleozoic APWP (screen); and, c) the B component paleopoles for the Oneota dolomites of Jackson and Van der Voo (1985; circle) and of this study (square) relative to the Ordovician reference poles (triangle). In b) and c), the poles are circumscribed by their ovals of 95% confidence.

6.2. The UMV fluid flow event

Economic mineralization was exploited within a core area of ~8000 km² in the UMV Zn–Pb district (Fig. 1). Heyl and West (1982) speculated that minor subeconomic occurrences of Zn–Pb mineralization in the surrounding ~100,000 km² were part of the UMV mineralizing event also.

Pannalal et al. (2004) reported paleomagnetic results from the core area of the UMV ore district, based on results from 33 sites of which 10 were in mineralization. They found that the ChRM of the 23 Middle Ordovician host rock sites resided mainly in pyrrhotite with magnetite being a carrier in only a few specimens. In their analysis, Pannalal et al. (2004) used end points only to obtain their site mean ChRM directions. Also, they excluded two site mean directions, one because it had poorly clustered specimen directions and the other because it gave an aberrant direction. The remaining 16 dolostone and 5 limestone sites carried a secondary chemical remanent remagnetization with slightly different paleopoles of 45.1°N, 126.7°E ($\delta p = 1.8°$, $\delta m = 3.5°$; Fig. 9b) and 43.5°N, 130.5°E ($\delta p = 3.2°$, $\delta m = 6.3°$), respectively.

6.3. A component

The paleopole for the A component is located close to the Early Permian portion of the North American APWP and overlaps the paleopole for Upper Mississippi Valley MVT mineralization and ore-stage host dolomites of Pannalal et al. (2004; Fig. 9b). The Early Permian age is consistent with pyrrhotite being the major carrier of both the A ChRM in this study and the Upper Mississippi Valley MVT mineralization and dolomites (Pannalal et al., 2004). The fact that the A paleopole is located a little to the east of the APWP (Fig. 9b) is attributed to trace remanence contributions to the ChRM of either: 1) primary Lower Ordovician magnetite, and/or 2) secondary hematite and/or goethite from recent weathering. Regardless, the A ChRM of the Oneota dolomite is deemed to be a secondary chemical remanent magnetization that was caused by the same fluid flow event that formed the Upper Mississippi Valley ore deposits and dolomitized their Middle Ordovician host rocks.

6.4. Boundary of the UMV fluid flow event

When the percentage of specimens with "A component end points" to all "measured specimens per site" is calculated for the 23 host rock sites of Pannalal et al. (2004), the percentages are all ≥40% except for their site 21 where the percentage drops to 33%. Conversely, Pannalal et al. (2004) report no clustering of specimen directions, let alone a site mean direction, in the primary Early Ordovician direction. Together, these observations suggest that a percentage of ≥40% for the "specimens with A component end points" to all "measured specimens per site" is a reasonable lower limit and outer boundary for the UMV fluid flow event because >95% of the carbonate specimens within the boundary retain Early Permian remagnetization end points. Fig. 10a shows the measured percentages for the 25 sites examined in this study and the determined 40% lower limit for the UMV district.

Sites 1 to 8 (Figs. 3 and 10b) were sampled to provide a profile extending ~60 km northeasterly from the northeastern corner of the core area of the UMV Zn–Pb district. The profile extends past Madison, Wisconsin, about to where Heyl and West (1982) drew their postulated outer limit for the UMV's subeconomic mineralization. With the marginal exception of site 5, it is evident that the Oneota dolomite has been substantially remagnetized by Early Permian fluids along the entire profile (Fig. 10a). A sidelight of this study is that the Oneota dolomite sample that gave a $^{238}U/^{206}Pb$ age of 504 ± 14 Ma (Winter and Johnson, 1995) was collected close to site 1, which exhibits thorough Early Permian remagnetization. The validity of the 504 Ma age is questionable because it is also significantly older at 95% confidence than the Oneota's stratigraphic age of 486 ± 3 Ma (Gradstein et al., 2004). Further, sites 9 to 16 (Figs. 3 and 10b) were sampled around the eastern flank of the UMV core area. Here, the east–west profile of Oneota sites 13 to 16 all show high percentages of A component end points that indicate substantial Early Permian remagnetization (Fig. 10b), but note that the two most easterly sites, 13 and 14, each yielded one specimen that retains a primary Ordovician B component (Table 1). Conversely, in the north–south profile of sites 9 to 12, site 9 displays a weak concentration of B

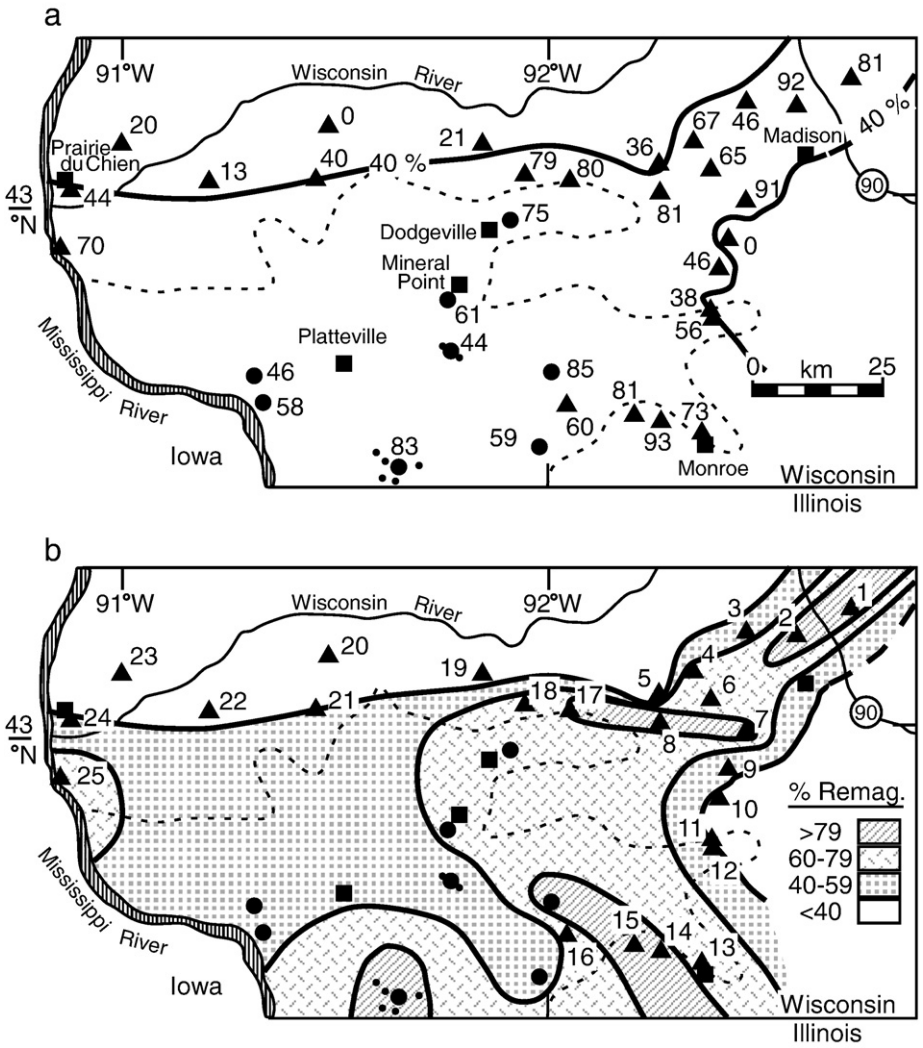


Fig. 10. Plots of the percentage of specimens with "A component end points" relative to the total number of specimens measured for each Lower Ordovician Oneota site analyzed in this study (▲) and for the core area of the UMV district (●) within the dashed line from Pannalal et al. (2004). a) The 40% contour (heavy solid line) for the site percentage indicates the outer limit of the UMV district based on the dominance of the Early Permian remagnetization in the Oneota dolomite. b) The site percentages of A component remagnetization contoured into quartiles with site numbers for this study.

component directions only, sites 10 and 11 retain marginal percentages of both A and B component end point directions, and only site 12 provides a preponderance of A component directions. These results indicate that sites 9 to 12 are straddling the outer limit of the UMV fluid flow event and that sites 13 and 14 may be approaching the outer limit.

Sites 18 to 25 provide a band of sites across the northern side of the UMV core area (Figs. 3 and 10b). Three of these sites (17, 18 and 25) have high percentages of A component end points although two (18 and 25) do retain two specimens each with B component end points. Four of the sites (19, 20, 22 and 23), conversely, have nil to low percentages of A component end points but do retain a strong to total preponderance of B component end points, and have clearly been little affected by the Early Permian fluid flow event. On the northwesterly profile of Jackson and Van der Voo (1985) in which 10 Oneota dolomite sites yielded a B component, their southernmost sites (W1 and W2) were collected close to site 24 of this study. Finally, sites 21 and 24 both gave marginal percentages in this study, suggesting that they are close to the outer limit of the UMV fluid flow event. From this band of sites in the Oneota dolomite, it is possible to draw an Early Permian "remagnetization contact" across the north side of the UMV district (Fig. 10a). This result indicates that the minor occurrences of MVT mineralization found to the north of Prairie du Chien along the Mississippi River valley on the west side of the Wisconsin Arch were unlikely to have been deposited by the UMV fluid flow event as Heyl and West (1982) speculated, but rather formed during diagenesis so that the primary ChRM of the Oneota dolomite was not remagnetized.

6.5. Source of UMV mineralizing fluids

In contrast to the minor MVT occurrences north of Prairie du Chien, the fluid flow that emplaced significant Zn–Pb mineral deposits in the UMV district also remagnetized the Oneota dolomite, thereby providing a sulphide tracer for the mineralizing fluids by pyrrhotite authigenesis. The results of this study show that the fluids did not originate from the north along the Wisconsin Arch, nor, when combined with the results of Jackson and Van der Voo (1985), from the northwest direction along the upper Mississippi river. Towards the northeast and southeast are two corridors aligned with sites 1 to 8 and 13 to 16, respectively, with high levels of remagnetization that could be recording fluid flow out of the Michigan Basin in the Early Permian (Fig. 10). In their study of the stratigraphy of the Prairie du Chien Group, Smith et al. (1993) have shown from isopach analysis that the group suffered either extensive dissolution or erosion prior to deposition of the overlying St. Peter sandstone along the same two corridors. Karsting from dissolution could have provided easy access

to facilitate a mineralizing fluid flow from the Michigan Basin as suggested by Bartos (2001). Note also that the percentages of remagnetization are highest in the southwest part of the study area where most of the economic Zn–Pb mineralization has been found, suggesting that additional economic mineralization may be present along the "highs" of the northeast and southeast corridors (Fig. 10b).

Testing for fluid flow from the Illinois and/or Forest City basins (e.g. Rowan and de Marsily, 2001) will require further paleomagnetic work along the southern and southwestern sides of the UMV district. At present, the relatively low first-quartile 55% rate of remagnetization in the cluster of sites 1 to 6 of Pannalal et al. (2004) on the southwest boundary of the core area of the UMV Zn–Pb district (Fig. 3 and 10a) suggests that the Illinois Basin is a more likely fluid source than the Forest City Basin.

The suggestion of Spirakis and Heyl (1996) that the fluid flow event was caused by convection from a high heat-producing Precambrian granite is unlikely for two reasons. First, there is no geophysical reason from magnetic or gravity surveys to believe that such a granite is centred under the UMV district (Mudrey, 2002), and second, this genetic model becomes progressively more unlikely as the Early Permian remagnetization signature shows expansion of the area affected by the UMV fluid flow event.

7. Conclusions

This study has shown for the first time that the boundary of a regional mineralizing fluid flow event can be successfully mapped using paleomagnetic methods. It has been shown that the Lower Ordovician Oneota dolomite carries either a primary B remanence or a secondary Early Permian A remanence component that is co-eval with the remanence of the Upper Mississippi Valley (UMV) Zn–Pb ores and adjacent host rocks. The boundary between A and B remanences can be defined by the percentage of specimens at a site that yield the secondary remagnetization remanence relative to the number of specimens tested. In this study, the boundary was tracked for >200 km across the northern and eastern sides of the UMV ore district. In addition, the same percentage calculation provides a crude measure of the intensity of conversion of iron oxides (magnetite) to iron sulphides (pyrrhotite), thereby indicating the potential for locating economic mineralization within the UMV ore district. At the local scale of adjacent sites, the boundary between the two types of remanence is sharp, and at regional scale the boundary is only a few kilometers wide at most. The reason for the location of the alteration boundary is not obvious but it appears to be related in part to prior karstification in the Oneota Formation shortly after deposition and in part to the geometry of the Wisconsin Arch. Finally, the fact that the Oneota dolomite underlies the Middle Ordovician carbonates hosting the UMV ore provides additional evidence for a process of ore formation in which the mineralizing fluids ascended from the underlying Cambrian sandstones and possibly also from the Precambrian basement.

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